

ASSESSING THE ROLE OF CHEMORECEPTION IN PARASITE AVOIDANCE IN THE
ACUTE BLADDER SNAIL *Physa acuta* FROM SUNSET POND IN SAN ANGELO,
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ABSTRACT

Freshwater snails such as *Physa acuta* act as intermediate hosts for a variety of parasitic flatworms in the class Trematoda, incurring high fitness costs due to their exploitation. Due to the long coevolutionary history between parasite and host as well as selective pressure to avoid parasites in their environment, this experiment sought to examine the ability of *P. acuta* to detect parasites or infected conspecifics using chemoreception, which is a freshwater snail's primary sense to observe their environment. *Physa acuta* has been repeatedly shown to have a higher parasitic load compared to other freshwater snail species existing in similar environments, so it was hypothesized that *P. acuta* would be unable to consistently detect parasites in their environment. To test this hypothesis, uninfected and prepatent-infected *P. acuta* were placed in choice chambers with one chamber containing water taken from wells containing infected conspecifics and the other chamber containing water taken from wells containing uninfected conspecifics. Snails were then observed to determine avoidance of the side containing water from infected conspecifics. Naïve snails raised in captivity as well as wild caught snails were used to identify any response consistent across the species as opposed to responses specific to the environment from which the snails were obtained. Binomial tests and logistic regression were performed in R and used to identify the significance of any avoidance behaviors as well as what metrics may affect detection rate, such as age, parasite species, or prepatent infection status. Results indicate that there is no significant, innate parasite avoidance response in *P. acuta*. However, infected snails still in the prepatent stage of infection showed consistently higher levels of detection amongst all snails tested. These results indicate the possibility of a learned, adaptive response post-infection in *P. acuta* which may reduce parasite load and risk of secondary infections.

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INTRODUCTION

Freshwater ecosystems are vitally important for maintaining biodiversity on the planet. While freshwater systems cover less than 1% of the Earth's surface, they contain over 10% of the Earth's known biodiversity (Dudgeon, 2006). Monitoring the health and activity of these diverse freshwater environments is incredibly important in the study of ecology and conservation as biodiversity is known to be a key factor in biomass productivity and health of an environment (Duffy, 2017). Understanding how organisms interact with one another in their environment is vital to understanding and maintaining this biodiversity, and if we can understand these interactions then we can better protect vulnerable species and ecosystems. The monitoring of invasive species in these freshwater systems additionally provides merit due to understanding factors that may lead to greater dispersion or competitiveness in an environment.

Freshwater snails (Class Gastropoda) play several important ecological roles in aquatic ecosystems. Generally considered herbivores or omnivores, these gastropods will scrape food with chitinous teeth attached to a muscular tongue known as a radula. Along with other members of the benthic community, these snails play a key role in consumption and control of invasive epiphytic algae populations (Brönmark, 1998). Algal consumption from snail populations has also been shown to induce growth of submerged plant life, leading to greater biodiversity in environments containing these organisms (Yang, 2020).

Freshwater snails can also act as ecological indicator species due to their sensitivity to synthetic chemicals and waste runoff, and by selectively choosing aquatic habitats with greater dissolved oxygen content (Wu, 2017). This habitat selectiveness shows that the presence of freshwater snails can be used to indicate the relative health of an aquatic ecosystem (Nurinsiyah, 2016).

In addition to promoting ecosystem health through feeding habits, snails also serve as important food sources for a variety of aquatic and semi-aquatic predators. Arthropods (such as crayfish and crabs), nematodes, reptiles, birds, mammals, and many species of fish will all feed on these small invertebrates. Some species, such as the snail kite (*Rostrhamus sociabilis*), have adopted a diet almost exclusively of apple snails, *Pomacea* spp. (Sykes, 1987). With such a large array of predators from many different phyla, snails have evolved a variety of antipredator tactics and behaviors. These include passive antipredator defenses such as altering shell morphology and development in order to better defend against local predators (Hoso, 2008) as well as active antipredator behaviors such as shell-shaking, burrowing, and floating into the water column to avoid benthic predators (Wethington, 2018). Considerable research has been done on these antipredator behaviors and adaptations in these snails, however, there is little research done on antiparasitic or parasite avoidant behaviors and adaptations in these organisms (Buck et al., 2018).

Parasitism is a form of symbiosis characterized by one of the participants, the parasite, harming or living at the expense of another organism, the host (Roberts et al., 2012). During this exploitation, the parasite will live on or inside the host. Notably, many parasites utilize several hosts over their life cycle, and each host will be harmed by the parasite. Parasitism has evolved independently many times across a wide variety of taxa including

animals, plants, fungi, protists, and bacteria. There are in fact more known species of parasitic organisms compared to nonparasitic organisms in the world (Roberts et al., 2012). The relationship between parasite and host is often very intricate and specific, as certain parasites will only infect a single species or similar, less inclusive taxonomic level of organisms because of the parasite's physiology, the host's physiology, environmental restrictions, or host immune responses.

Many parasitic organisms have intricate, multi-stage lifecycles that may require transition to and from one or more hosts (Figure 1). The phylum Platyhelminthes includes parasitic flatworms including the digenetic trematodes (Class Trematoda, Subclass Digenea). All trematodes exploit molluscs, such as snails and bivalves, as the first intermediate host in their multi-stage lifecycle. In this host, the parasite will reproduce asexually while consuming or absorbing host tissue.

Trematode parasites have evolved several routes of infection into the snail first intermediate host. One such route of infection can occur when a snail consumes embryonated trematode eggs in their environment. Once consumed, a miracidium hatches from the egg and penetrates into the hepatopancreas of the snail host. These miracidia are a ciliated larval stage that develops into a separate life cycle stage known as a sporocyst. In the second route of infection, the egg hatches in the water and releases the free-swimming miracidium that finds and directly penetrates the snail host to infect them. The miracidium then transforms into a sporocyst if the species of snail is a suitable host. These larger sporocysts exist within the host, reproduce asexually, and ultimately produce more sporocysts or a separate life cycle stage called redia. Rediae also reproduce asexually, but they acquire nutrients by directly consuming host tissue.

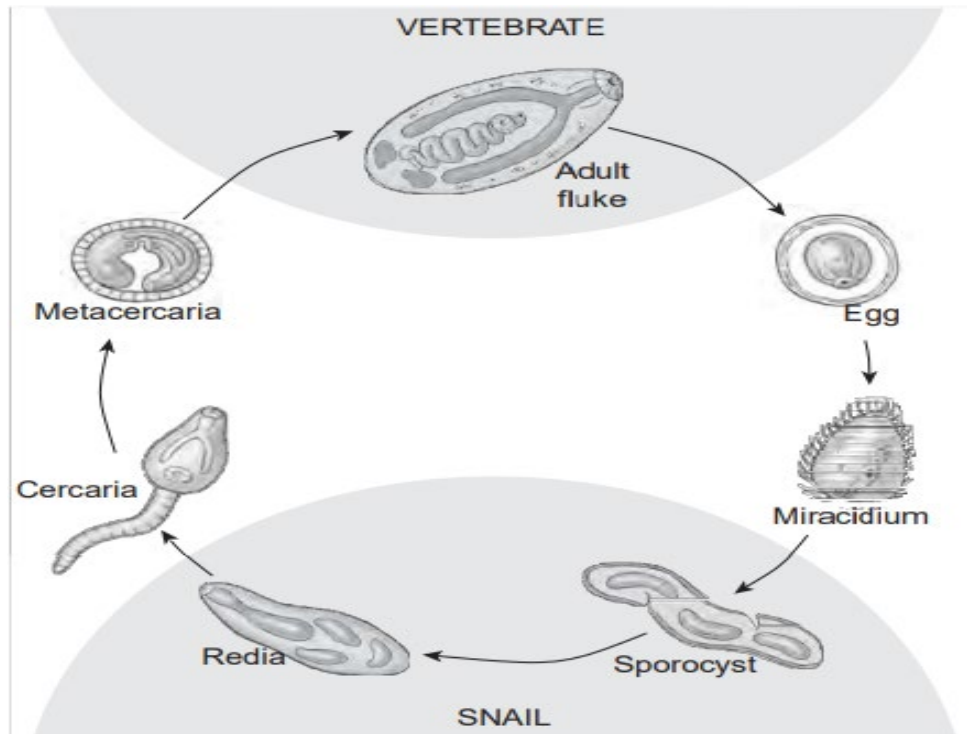


Figure 1: Generalized trematode life cycle. Sporocysts and rediae reproduce asexually inside this first intermediate host. Image from Roberts et al. (2012).

Cercaria are the next larval stage that will exit the snail and infect the next host. The route of infection into the snail and the pattern of intramolluscan stages is parasite-species specific with cercariae being produced by either sporocysts or rediae. Free-living cercariae can infect other organisms and develop into a metacercaria (larval stage that acts as a resting stage) or an adult worm. If the host is a suitable definitive host, then the parasite will mature and reproduce sexually within that host. In nearly all cases, vertebrates, such as herpetiles, birds, fishes, and mammals, are the definitive hosts for digenetic trematodes.

During their symbiosis with the snail host, sporocysts and redia will absorb or consume host tissue, respectively. Infection typically begins in the hepatopancreas and as the trematode population size increases, the larval stages spill over into the high-nutrient gonadal

tissue and consume it, thus castrating the snail host (Esch, 1994). This relationship between host and parasite has been ongoing for millions of years, and now, most trematodes require the presence of gastropods in their environment to complete their reproductive cycle.

Exploitation of the gastropod hosts incurs high fitness costs such as reduced fecundity, interruptions to sexual maturity, and increased mortality rates (Esch, 1994). Due to the coevolutionary history between these two groups of organisms as well as increased fitness costs associated with parasitic infection, there exists a possibility that gastropods have evolved a mechanism of avoiding these trematodes in their environment.

Without lenses in their eyes, gastropod vision is considered poor at creating a focused image and is mainly used for detecting the general presence of light rather than differentiation of objects or organisms in the environment (Zieger, 2008). Without a developed sense of sight, other senses must be utilized in order to detect food, mates, or danger, such as predators, parasites, or harmful substances.

Chemoreception is the process of detecting chemical stimuli in an organism's environment, and most chemoreception in animals is done via a sense of olfaction or gustation. Many gastropods will use chemoreception via well-developed olfactory organs at the tips of each tentacle as the primary way to perceive objects, especially at a distance (Chase, 2001). Gastropods have been known to use chemical signals for a variety reasons, such as sending or receiving mating signals and triggering the formation of egg-laying clusters (Cummins et al., 2005), predator avoidance (Jacobson, 2004), or nutrient foraging (Murphy, 2001). In aquatic ecosystems, it is common for snails and other small prey animals to assess the risk of predation via these chemical signals (Chivers and Smith, 1998). It is

currently unknown, however, whether these animals can utilize chemoreception to detect parasites in their environment in a comparable manner to predator detection.

Pulmonate freshwater snails are highly social animals forming population clusters or aggregates to feed, forage, and reproduce, even in hermaphroditic species (Simpson 1973). These population aggregates increase the risk of contact with infective trematode species leading to higher infection rates but reducing parasite load overall due to lower risk of multiple infections within a single snail host (Buck, 2017). The social nature of freshwater snails as well as this increased risk of infection suggest that any parasite avoidant behavior may lead to a fitness benefit, especially those behaviors that include identifying and avoiding infected individuals or social aggregates.

Physa acuta is a globally invasive, pulmonate, freshwater snail native to the ponds and lakes of North America. These animals are extremely invasive and have colonized both natural and artificial waterways around the world. They are also voracious and exhibit a fast rate of reproduction, thus outcompeting many other species of freshwater snails (Lydeard, 2016). Pulmonate freshwater snails are characterized by a respiratory system consisting of a single lung-like pulmonary cavity located inside the mantle underneath the shell. *Physa acuta* has modified this pulmonary cavity into a characteristic “air bladder” which may be used to store oxygen for extended periods of time while underwater, but it also is beneficial in dispersal because the air bladder allows the snail to float on the water’s surface. Additionally, these snails have been shown to utilize waterfowl as a dispersal vector, allowing for greater potential to invade waterways (Van Leeuwen, 2012). These varied routes of dispersal combined with a high tolerance for salinity and water pollutants (Kefford and Nugenoda, 2005; Turner and Montgomery, 2009) increase their invasive potential across the globe.

Physa acuta are known to be large consumers of aquatic plants and algae in their environment, which necessitates higher motility compared to other snail species. This increased motility leads to higher exposure to the parasite infective stages in their environment and higher parasitic infection rates compared to other potential freshwater snail hosts, especially when comparing infections of parasites with free-swimming infective larval stages (Zimmermann et al., 2014). This higher rate of trematode infection has led *P. acuta* to be associated with an increased spread of various parasitic diseases, both human and non-human (Dumidae et al., 2024). Therefore, examining the role of parasite avoidance by the potential global disease vector *P. acuta* should be a key area of study for malacologists and parasitologists alike. With those factors in mind, this study aims to identify if *P. acuta* exhibits parasite avoidance behaviors when detecting infection cues in water.

METHODS

Site selection

Sunset Pond in San Angelo, Texas, is a well-studied environment with a large population of *P. acuta*, *Helisoma anceps*, and *Gyraulus* sp. snails, as well as a large number of helminth parasites. Previous studies identified parasites from several families of trematodes, including Echinostomatidae and Strigeidae, both known to be harmful to the reproductive success of infected hosts, and both having varied infection patterns in invertebrate intermediate hosts. Due to its proximity to Angelo State University and as being a well-studied environment with consistent populations of both parasites and snails, Sunset Pond was the sole collection site of all specimens. In order to maximize diversity of both snails and parasites, snails were collected from three locations in the pond (Figure 2).



Figure 2: Sunset Pond with collection points (white marks) in San Angelo, Texas.

Specimen Collection

During May-November of 2024 and February-March of 2025, *P. acuta* and *H. anceps* were collected opportunistically from Sunset Pond at one of three collection sites (Figure 2). As both snail species inhabit freshwater lakes and ponds where they consume and graze upon the aufwuchs in their environment, collections were carried out via sifting through surface-level plants and algae, and collecting any attached snails. Snails were stored in containers filled with pond water to prevent desiccation or added stress. Species identification was carried out visually in the field.

Snails captured from Sunset Pond were then isolated in individual wells containing 10 ml of aged, dechlorinated, aerated tap water and given food from a high nutrient solid diet created by mixing 40 ml of water, 1 g agar, and 5 g crushed Tetramin fish flakes (#1393413, Tetra Werke, Melle, Germany) (Keas and Esch, 1997). During this period, 3 *P. acuta* egg sacs were also obtained from collected vegetation and isolated in a 60 cm x 30 cm x 45 cm aquarium in order to raise a separate population of snails free of interspecific competition and risk of parasitic infection.

Parasite Identification

Parasite infections in the snails will be patent or non-patent. In patent infections, the larval stage population is sufficiently high that the intramolluscan stages begin producing cercariae that escape from the host (Dreyfuss, 1994). To identify patent infections, all snails captured from Sunset Pond were observed for a 24-hr period to identify all that were shedding cercariae. Initial, visual identification of patent infection was performed under an Olympus SZ51 dissection scope (Olympus Global, Westborough, MA). If a patent infection was observed, trematode identification was performed by removing a small amount of

cercariae from the infected well and staining the sample with neutral red dye to improve visualization (Frandsen and Christensen, 1984). Stained cercaria were then visualized under greater magnification using an Olympus BX53 microscope (Olympus Global, Westborough, MA) to identify the specific cercaria types based on morphology. Cercaria were identified morphologically to the family level using Schell (1985).

Snails with non-patent infections still have larval stages reproducing asexually, but the population size has not reached the threshold when cercariae are produced. As such, trematode-infected snails may not be actively shedding cercaria during time of experimentation. Therefore, all trial snails were crushed post-trial to identify pre-patent infections. Parasites observed in the snail tissue were identified (Schell, 1985) and the snails infection status was recorded.

Objective 1: Can snails sense food cues?

Chemoreception in *P. acuta* was first tested by determining if snails can sense and move toward a food source. Trials were conducted using 40 wild-caught and uninfected snails that were isolated and starved for 7 d following an initial feeding of a solid, high nutrient diet. The solid diet and a separate, liquid form of this diet was tested in this experiment. The liquid diet lacked agar and was created using 5 g Tetramin fish flakes dissolved in 40 ml water.

Post starvation, snails were separated into 2 trial groups: one with a solid diet and one with a liquid diet. Snails in the liquid diet trial were placed in the center chamber of a 22 cm x 10 cm choice chamber (#746619, Carolina Biological Supply, Burlington, NC) containing 50 ml aerated tap water along with 0.50 ml of the liquid diet pipetted into the center of one side of the chamber. The chamber that received the food was alternated for each trial to

remove any bias in snail preference for side of the choice chamber. In the trials using the solid diet, snails were placed in the center of a Carolina choice chamber containing 50 ml aerated tap water, and 0.5 g of the solid diet was added to the center of one side of the chamber. Similar to the liquid diet trials, the side of the choice chamber receiving the test condition alternated between the right and left side for each trial. Movement patterns and time until movement into either chamber were observed to determine detection of the solid or liquid diet. Positive detection was determined only if the trial snail had fully entered the chamber containing the diet. Each snail was tested once in this experiment.

A secondary test was conducted to determine the rate that a chemical signal would become ubiquitous in both chambers. For this test, 0.5 ml red food coloring was pipetted into the center of one of the large side chambers in a choice chamber containing 50 ml aerated tap water. The time of diffusion of the red food coloring into the opposite chamber was recorded. This time served as reference for the maximum time allowed for snails to choose between the two larger chambers.



Figure 3: Carolina choice chamber. Snails were placed in the middle compartment perpendicular to the openings to the two larger chambers so that they can assess and choose one of the two larger chambers. Snail choice was determined once the snail completely entered either larger chamber

Objective 2: Can snails identify a parasitized environment?

To determine the ability of *P. acuta* to identify and react to environments containing larval trematodes or conspecifics infected by larval trematodes, trials were performed in a laboratory setting using simulated environments of infected or uninfected waters.

Water samples were obtained from wells containing infected *P. acuta* that had been isolated for 7 d to increase the concentration of any warning cues that may be found in the water. To simulate parasite-free environments, water samples were obtained from wells containing isolated *P. acuta* not shedding cercaria. All snails used to simulate parasite-free waters were crushed prior to use in order to identify any patent infections that may have released chemical cues into the water.

Sunset Pond is host to a diverse trematode population, and snails may react differently based on different parasites. Therefore, water samples obtained from infected snails were classified based on the family of parasite infecting that snail. All water samples were kept at room temperature and used within 2 h of collection from wells.

For each parasite family observed during the collecting period, 20 snails were selected based on the absence of shed cercaria following 7 d of isolation. These trial snails were then placed individually in a Carolina choice chamber containing 50 ml aerated tap water along with test conditions of 0.50 ml of water isolated from infected conspecifics pipetted into one chamber and 0.50 ml of water obtained from uninfected conspecifics pipetted into the other. Trial snails were placed so that the aperture of the shell was equidistant from both chambers and the snail initially faced perpendicular to the large chamber openings. Movement was recorded until the trial snail fully entered either chamber. Additionally, if no chamber was entered by the end of the 3-min observation period, then the trial was marked as a no detection response. Each trial snail was tested 5 times using a new chamber for each replicate. The chambers containing infected or uninfected water were reversed in order to account for any directional bias exhibited by the trial snails. Upon completion of each trial, the trial snail's shell length was measured from aperture to the apex of the shell spire. This measurement would function as an indicator of the snail's age. Additionally, trial snails were crushed to identify any patent infections, and these infections were recorded along with their results.

Objective 3: Are behavioral avoidance mechanisms an innate response?

There are several possibilities as to the origin of any parasite avoidance response, so this objective sought to determine if avoidance responses are an innate response. Lab-reared

P. acuta were used because they were raised in a parasite-free environment. *Physa acuta* were reared for a minimum of 13 wk, which is how long it takes these snails to reach sexual maturity (Nuñez, 2010). The experimental procedures above were repeated using these lab-reared snails as the trial snails.

Statistical Analysis

All data were analyzed using the open-source computer programming language and software environment R, version 4.4.2 (R Core Team, 2024), and the integrated development studio RStudio (Posit, 2023). Origin of the snail, shell length, and prepatent infection status were included as factors when analyzing the data. Choice exhibited by the trial snails was recorded dichotomously for each trial. Five replicates were performed for each trial snail, so parasite avoidance was recorded as a frequency: 0 indicates that the snail avoided the parasitic environment in none of the 5 replicates, and 1 indicates that the snail avoided the parasitic environment in all 5 replicates.

A one-tailed binomial test was used to determine if snails exhibited preference by moving toward a food source. For the parasite-avoidance experiments, mixed-effects logistic regression analysis was performed to examine the effects of snail shell length, snail origin, infection status, and trial parasite type on the binary choice response while accounting for individual snails as a random effect. To do this, a generalized linear mixed-effects model with a binomial distribution was created using the package lme4 (Bates, 2015). The fixed effects of the model consisted of origin (wild vs. lab reared), status (infected vs. uninfected), shell length (continuous), and trial parasite based on cercariotype (*Strigea* vs. *Echinostome*). A random intercept was included for individual snail ID to account for the fact that 5 replicate trials were performed for each snail. Model simplification was performed based on

Akaike Information Criterion (AIC) to identify the simplest or most parsimonious logistic regression model that explains the most variation in the response variable (Anderson and Burnham, 1999).

RESULTS

Parasites Identified

Over the sampling period, 2 distinct cercarial morphologies were observed infecting the snails: *Strigea* and *Echinostome*. This indicates that snails were infected by trematodes belonging to the Families Strigeidae and Echinostomatidae, respectively (Fig. 4; Schell, 1985). Both parasite morphologies observed were used in parasite avoidance experiments for objectives 2 and 3.

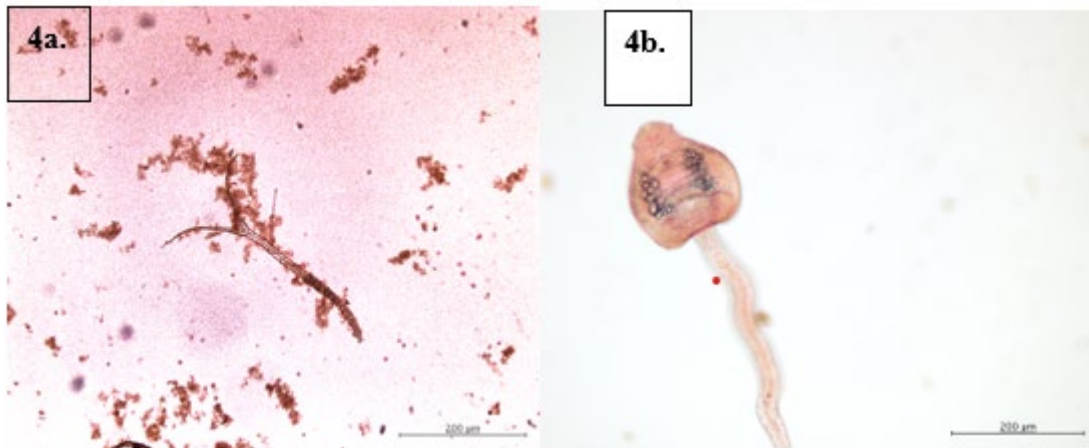


Figure 4: Cercaria identified from *Physa acuta*. 4a.: *Strigea* cercaria. 4b: *Echinostome* cercaria.

Snail Behaviors

Snails showed significant difference in detection response rate in the food trials. In the solid diet trials, 9 of 20 *P. acuta* identified and approached the chamber that contained the food, indicating no significant level of detection (Binomial test: $P=0.75$). However, 18 of 20 *P. acuta* were able to identify and approach the chamber with the liquid food, indicating a significant level of detection (Binomial test: $P=0.0002$). The mean result of the dye tests performed was approximately 3.5 min. Since all positive detections in the liquid trials

occurred within 3 minutes of exposure, the time period for observation in the parasite trials was limited to 3 minutes post-exposure to the test conditions.

Overall mean response rate among the uninfected snails in the parasite trials for both cercariotypes was 0.48. Of the 40 wild-caught snails used in the experiments, 21 of these snails showed prepatent infections. These parasitized snails exhibited a heightened parasite avoidance response compared to unparasitized snails with 86% moving away from the side receiving infected snail water (Fig. 5). Eight of 10 infected wild-caught snails in the *Strigea* trials had avoidance rates of 80% or higher, and 10 of 11 infected wild-caught snails in the *Echinostome* trials had response rates of 80% or higher. Sample sizes differed because prepatent infection status could only be determined after the trials were completed.

The full model analyzed with mixed-effects logistic regression indicated that prepatent infection status had a significant effect on the probability of avoidance (Type II Wald Chi-square Test of Deviance: $P < 0.0001$). Specifically, uninfected snails were significantly less likely to avoid a parasitized environment compared to infected snails. Other predictors, in the model were not statistically significant ($P > 0.19$ for the remaining terms; Fig. 6).

The full mixed-effects logistic regression model generated an AIC of 519.4. The simplest model with the lowest AIC (516.0) included infection status and no other variable. Removing infection status from any model resulted in a large increase in AIC values. These results suggest that infection status of the trial snail was the most important variable, and the inclusion of other metrics did not meaningfully improve the model's explanatory power.

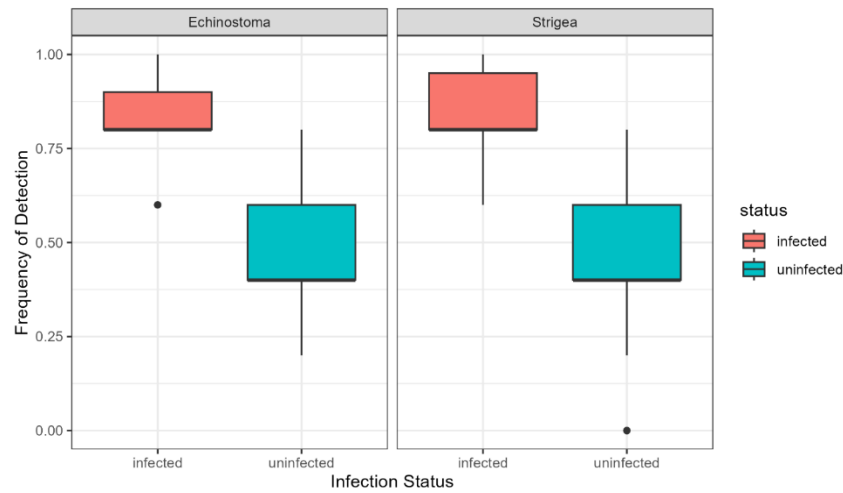


Figure 5: Mean frequency of parasite avoidance response for prepatent and uninfected snails across 5 replicates for each of the trematode infections. Mixed-effects logistic regression strongly suggests that infection status causes snails to avoid parasite-infested water ($P < 0.0001$).

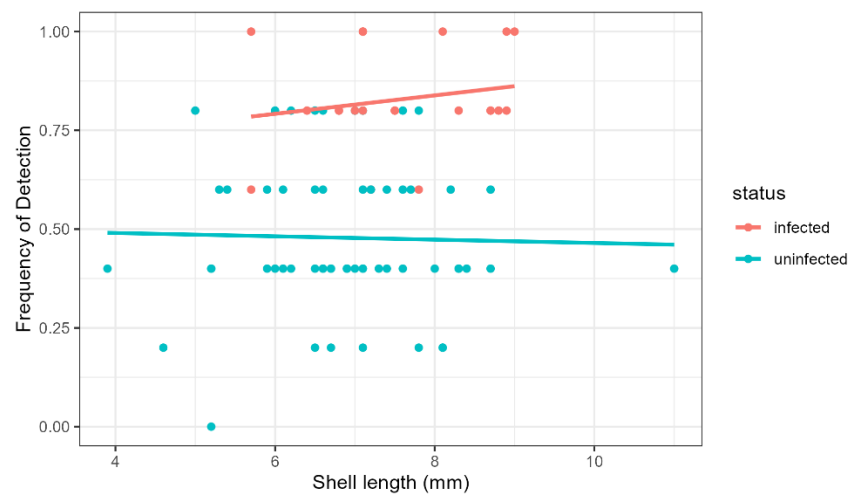


Figure 6: Frequency of parasite avoidance response in snails compared to snail shell length across 5 replicates, with mean response as a linear regression line for uninfected and pre-patent infected snails.

DISCUSSION

This experiment demonstrated that snails can sense environmental cues and respond accordingly. Prior to entry into either large chamber, trial snails were observed to consistently move in a circular motion, seemingly to assess their environment. Starved snails consistently move toward a food source. However, the type of food source has an effect on movement. Specifically, snails exhibit movement into the side receiving a liquid diet, but did not exhibit a preference when presented with a solid diet. This difference is likely due to the diffusion of chemical cues in the water: chemicals are free to move throughout the water when in liquid form but are bound to the agar in the solid source. Chemical cues from the solid diet are likely diffusing into the water, and given sufficient time, it is expected that starved snails will ultimately sense and move toward the food source. These results informed the methods of the parasite trials that followed. In particular, chemical cues present in the water that originate from parasites and infected conspecifics may be detected by other snails, and concentrated chemical cues, when added to the choice chambers, may elicit a response in the trial snails within 3 min.

Uninfected snails responded to a food stimulus but not to infected snail water. Neither shell length, parasite species tested, nor origin affected the mean response of the snails. Thus, it initially appears that infected snails or their parasites do not release warning chemicals into the water. Alternatively, chemical cues are produced by infected snails or their parasites, but conspecifics are unable to detect them. The final option is that chemical warning cues are released into the water and snails can detect them, but the cues are not associated with danger to the individual and are thus ignored. This third option appears most likely because prepatent infected snails exhibit strong avoidance behavior compared to uninfected snails.

The difference in response clearly indicates that infection leads to some behavioral shift within the host when they detect another infected individual. There are several reasons why this result occurred, and each has drastically different implications on the relationship between parasite and host.

One potential explanation is that snails learn to associate chemical signals shed by the parasite or infected host only after being infected. Snails and other gastropods have shown capability for learning (Alkon, 1984). For example, the pond snail *Lymnea stagnalis* has been experimentally shown to exhibit configural learning, and they form antipredator behaviors after associating multiple stimuli from food odors and predator effluent to danger (Kagan, 2019). Thus, there is potential that snails learn to avoid other infected individuals.

In this study, the route of infection for the two parasite families differs. Snail hosts shedding strigea cercariae were infected through passive means (the consumption of an egg). After consumption of an embryonated egg, a miracidium hatches from the egg and penetrates the gut mucosa of the host to reach the snail hepatopancreas, causing damage along its path (Pyrka, 2022). Snails shedding echinostome cercariae were infected through direct penetration of host tissue by a free-swimming larval miracidium. This causes immediate harm and host damage. Moreover, echinostome cercariae are produced by rediae in the snail host that actively consume host tissue and cause additional harm to the host. Because of the difference in infection pathway and harm caused by the parasites, it would be more likely that an immediate, strong response would develop following infection by members of the echinostomes compared to species with strigea cercariae, even though both cause considerable harm to the host. Future work is needed to determine when parasite avoidance

behaviors develop following infection and if this timing differs based on the specific parasitic infection.

A second alternative is that the infected snails are not responding to a signal at all but are being manipulated by their parasitic symbionts to ensure a competitive advantage over other parasites in the environment. While a parasite's environment may differ from other organisms in that all or most of their nutrients must come from their host, there is still competition that occurs when a host is infected with multiple parasites. Trematodes have been shown to exhibit antagonistic behaviors towards other trematodes in their environment (Lie, 1973). This antagonism can, in some cases, even lead to the development of nonreproducing, obligatory sterile "soldier castes" of rediae tasked with preventing competitors from accessing hosts (Metz, 2024). Antagonistic behaviors such as these are one reason infections by multiple species of trematodes are rarer than would normally be expected in any given environment (Kuris, 1994). Behavioral modification of a snail host to avoid trematode infested areas of the environment would be beneficial to any parasite already infecting the snail.

There are several potential research avenues arising from this research. First, experimental infection of snail hosts in the lab allows researchers to accurately identify when parasite avoidance behaviors initially develop and determine the length of their persistence. Secondly, analysis of additional trematode species would help determine if avoidance is universal across all trematode infections, and it may shed light on how the different infection pathways affect the development of avoidance behaviors. Third, additional host species should also be examined. There is a wide variety of snail species being exploited by digenetic trematodes, and freshwater snail species exhibit varying levels and speeds of mobility. Thus,

the ability to avoid these environments may be a limiting factor in this particular avoidance response. Finally, differences in occurrence rates of multiple infections should be assessed in snail hosts that exhibit this parasite avoidance behavior compared to any snail host species that do not exhibit this avoidance behavior.

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VITA

Alexander Bell graduated from San Angelo Central High School in June 2013. The following August he entered Angelo State University and subsequently left the following semester. After several years working and traveling as a competitive trading card game player, Alexander returned to Angelo State University in January 2019. In August 2022 he received a degree of Bachelor of Science in Biology. He then returned to Angelo State University again in August 2023 to pursue his Master of Science Degree in Biology. During his time pursuing his master's degree he worked as a graduate assistant for two semesters as well as a teaching assistant for an additional two semesters. Additionally, he presented research at two professional meetings. The first was at the Texas Academy of Science Annual Meeting 2025 where he presented "Habitat selection in the globular drop snail *Helicina orbiculata* in East Texas waterways using GIS data" as an oral presentation. The second was at the Southwestern Association of Parasitologists Annual Meeting 2025 AKA Casino SWAP 2.0, where he presented his thesis work "Assessing the role of chemoreception in parasite avoidance in the acute bladder snail *Physa acuta* from Sunset Pond in San Angelo, TX" as an oral presentation and received the Mark H. Dresden award for best graduate student presentation.